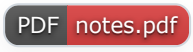
Notes_25.04: Classification Evaluation Metrics







🌐 **Language: English** | हिंदी / Hinglish | ⬅️ Lecture 25 Overview

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📌 Executive Summary

Accuracy paradox, Precision, Recall, Specificity, and the harmonic mean F1-Score.

🎯 Mathematical Formulations

Classification Metric Equations:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall (Sensitivity)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$$

💡 Conceptual Principles & Architecture

Accuracy Paradox (Accuracy Dhoka Kyu Deti Hai?):

Maan lijiye 1000 logo mein se sirf 5 logo ko cancer hai (99.5 negative class). Agar model bina soche sabhi ko 'Negative' bol de, toh model ki Accuracy 99.5 hogi!

Lekin asal mein cancer wale saare 5 mareez mar jayenge. Isliye Imbalanced data mein Accuracy useless hoti hai.

Precision vs Recall Kab Use Karein?

- **High Recall Zaroori:** Medical diagnosis (Cancer detection) ya Fraud detection mein. Hum ek bhi positive patient chhod nahi sakte (FN minimize karna hai).
- **High Precision Zaroori:** YouTube recommendations, Spam filters, ya Job applicant screening mein. Agar koi email genuine hai toh woh Spam folder mein nahi jaani chahiye (FP minimize karna hai).

💻 Python Production Implementation

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_scc

y_true = [1, 0, 1, 1, 0, 1, 0, 0, 1, 0]
y_pred = [1, 0, 1, 0, 0, 1, 1, 0, 1, 0]

print(f"Accuracy  : {accuracy_score(y_true, y_pred):.4f}")
print(f"Precision : {precision_score(y_true, y_pred):.4f}")
print(f"Recall    : {recall_score(y_true, y_pred):.4f}")
print(f"F1-Score  : {f1_score(y_true, y_pred):.4f}")
```